

Sales Data Cleaning & Analysis

InternSpark Business Analytics Internship — Task 1

Executive Summary

The analysis covers 2,823 order-line records and 307 unique orders from January 2003 through May 2005. Total recorded sales are \$10.03M, with an average order value of \$32,679.57. The analysis focuses on data quality, product performance, seasonality, geographic concentration, order status and actionable business recommendations.

Metric	Result
Total Sales	\$10,032,628.85
Unique Orders	307
Average Order Value	\$32,679.57
Dataset Period	Jan 2003 – May 2005

Key Findings

- **Core product:** Classic Cars contributes 39.07% of total sales. Classic Cars + Vintage Cars contribute 58.04%.
- **Growth:** Planes grew 84.63%, Motorcycles 51.13%, and Vintage Cars 40.01% from 2003 to 2004, compared with 18.69% for Classic Cars.
- **Seasonality:** Average November sales are \$1,059,443, about 4.75× April's average. November 2004 is the highest observed month at \$1,089,048.
- **Geography:** USA contributes 36.16% of sales; USA + Spain + France contribute 59.35%.
- **Order health:** 286 of 307 unique orders (93.16%) are shipped. Non-shipped statuses represent 7.39% of sales value; cancelled sales value is \$194,487.

Priority Recommendations

1. Protect Classic Cars availability and performance because it is the largest revenue contributor.
2. Scale high-growth categories selectively, especially Planes and Motorcycles, while monitoring whether growth persists.
3. Plan inventory, sales capacity and fulfillment ahead of the October–November seasonal peak.
4. Protect the major USA/Spain/France markets while testing targeted expansion in selected smaller markets.
5. Investigate cancellations, holds and disputes, prioritizing high-value exceptions and recurring root causes.

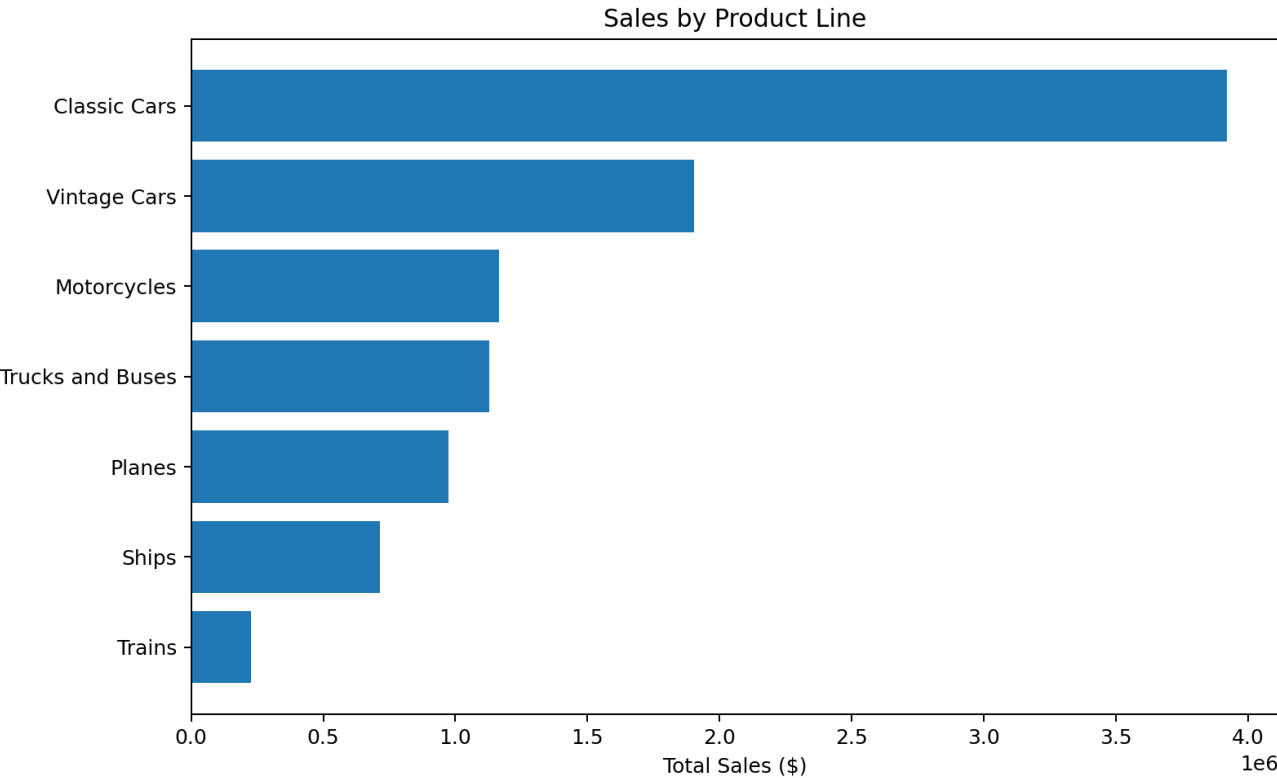
Data source & scope: Public Kaggle Sample Sales Data used as a representative internship dataset; recommendations are analytical and should be validated against current company data before implementation.

1. Data Quality & Cleaning

Quality Check	Result	Treatment
Rows / columns	2,823 / 25	Retained
Exact duplicates	0	No removal required
Invalid dates after parsing	0	ORDERDATE converted to datetime
ADDRESSLINE2 missing	2,521	Replaced with Not Provided
STATE missing	1,486	Replaced with Not Provided
POSTALCODE missing	76	Replaced with Not Provided
TERRITORY missing	1,074	Retained as missing; no unsupported imputation

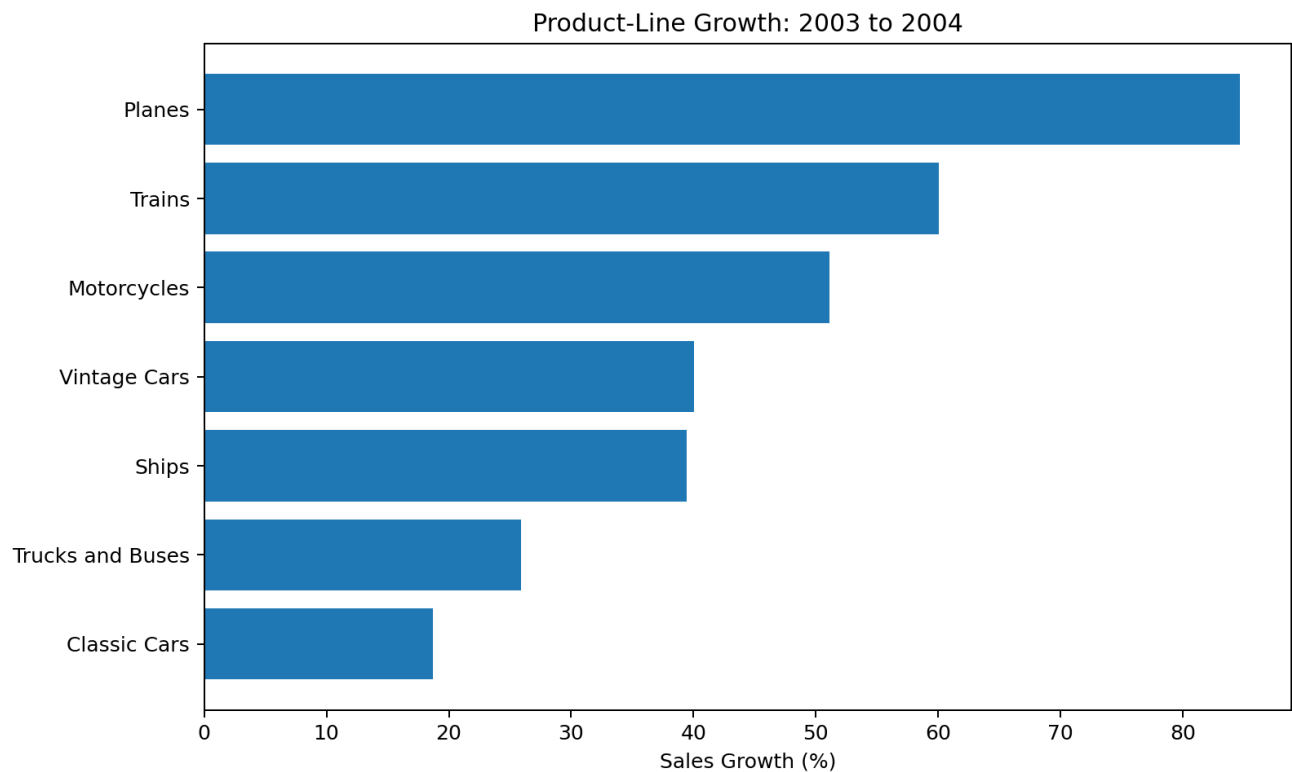
The source data contains 1,074 missing Territory values, concentrated in USA and Canada. Because the dataset does not provide a reliable mapping rule for those rows, Territory values were not fabricated.

2. Product Performance

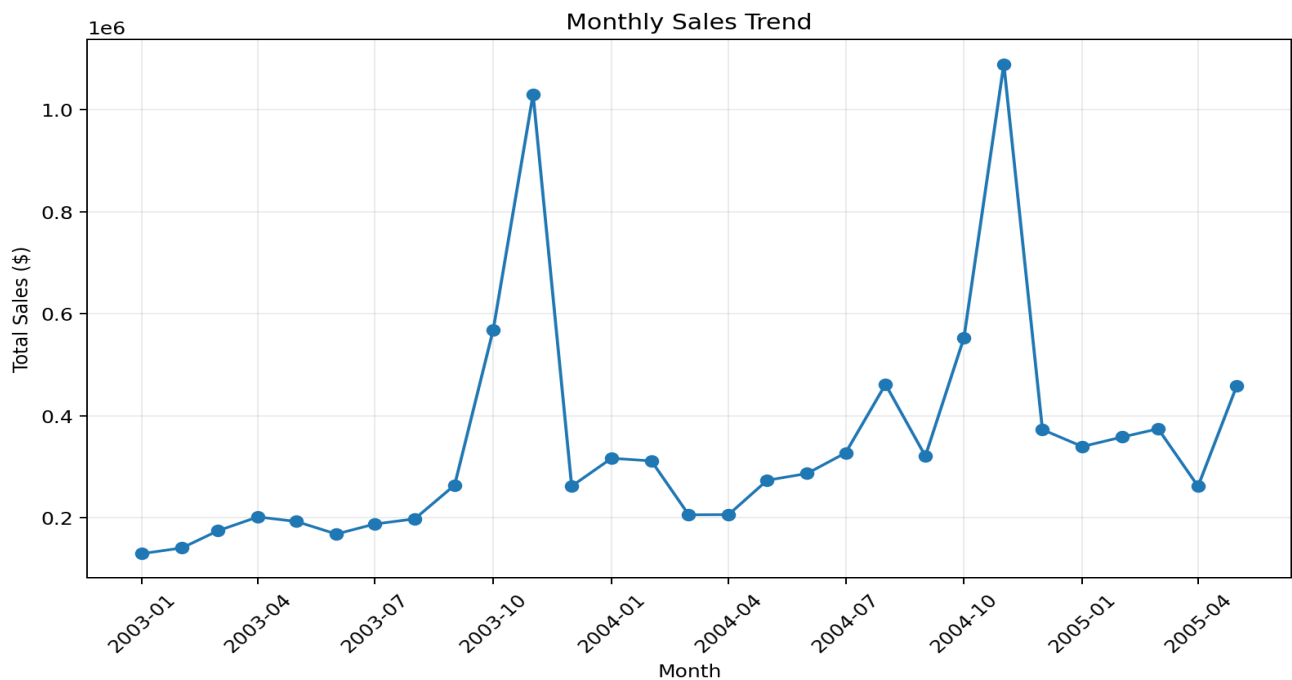


Classic Cars leads with \$3,919,616 (39.07% of total sales), followed by Vintage Cars at \$1,903,151.

3. Product Growth & Seasonality

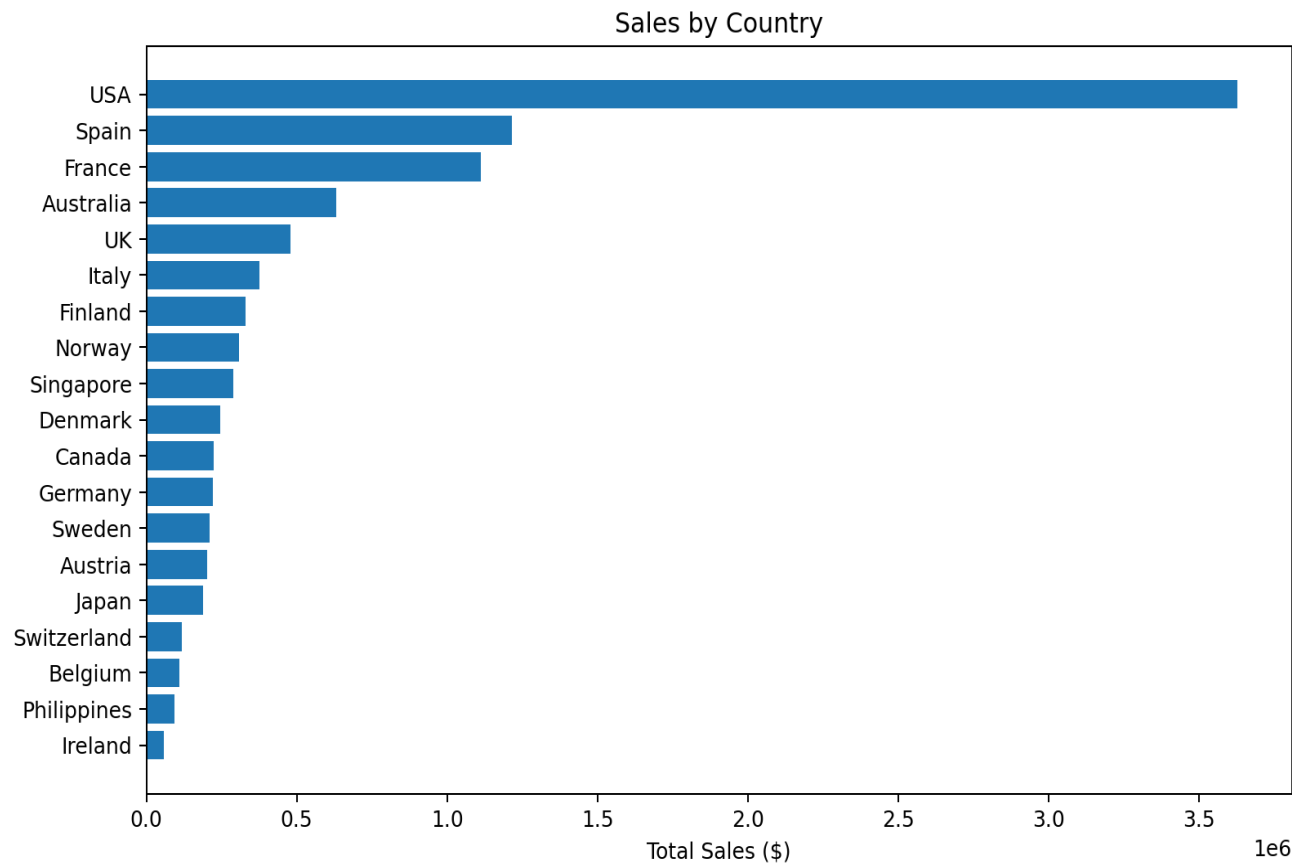


The 2003–2004 comparison shows that Planes had the fastest growth, followed by Trains and Motorcycles. Classic Cars remained the largest category but grew more slowly, supporting a strategy of protecting the core while selectively investing in faster-growing categories.



Sales show a recurring Q4 spike. Average November sales are \$1,059,443, versus \$223,130 in April. The highest observed month is November 2004 at \$1,089,048.

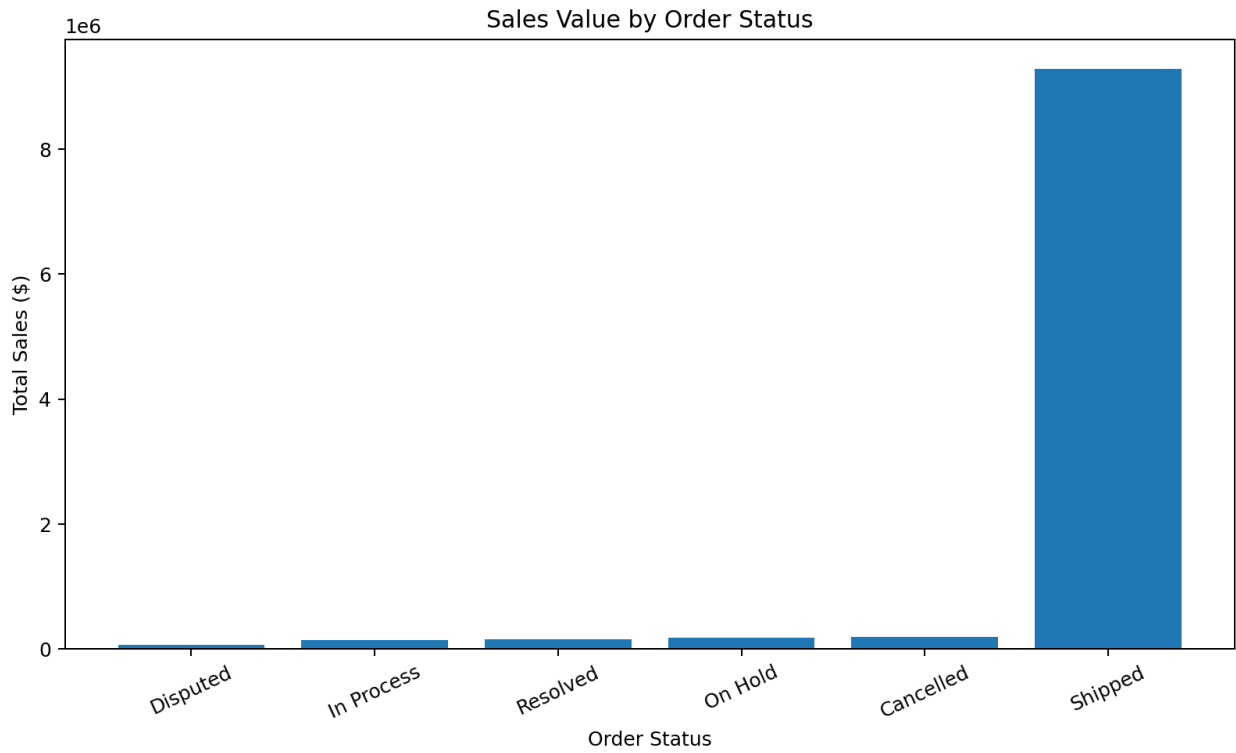
4. Geographic Performance



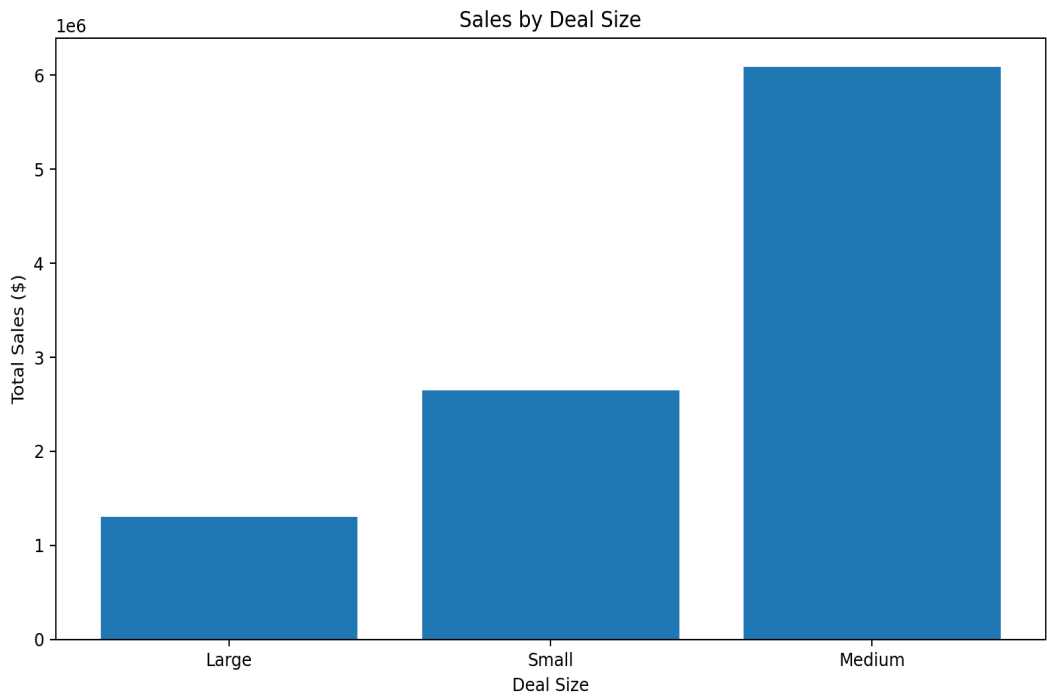
The USA generates \$3,627,983 (36.16%). Spain and France are the next two largest markets. Together, the top three markets contribute 59.35% of sales.

Country × product analysis also shows Classic Cars as a major contributor in the largest markets, while France has a comparatively strong Motorcycle contribution. This supports targeted rather than one-size-fits-all market actions.

5. Order Health & Deal-Size Mix



286 of 307 unique orders (93.16%) are shipped. Non-shipped statuses represent 7.39% of sales value. Cancelled orders represent 4 unique orders and \$194,487 of sales value.



Medium deal-size line items account for 60.68% of sales. Deal Size is not treated as an order-level share because 277 of 307 orders contain multiple deal sizes.

6. Recommendations & KPI Framework

Priority	Action	Evidence	KPI
1	Protect Classic Cars	39.07% of sales	Sales/share; stockouts
2	Scale high-growth categories	Planes +84.63%; Motorcycles +51.13%	YoY growth; sales share
3	Prepare for Q4 peak	Nov avg ≈ \$1.06M	Oct–Nov sales; fulfillment; stockouts
4	Strengthen major markets + test diverse categories	Top 8 = 59.35%	Country growth; contribution
5	Reduce operational exceptions	Non-shipped = 7.39% of sales value	Cancellation/dispute/on-hold rates; exception value

Assumptions & Limitations

- The dataset is at order-line level; ORDERNUMBER identifies unique orders.
- AOV is total sales divided by unique orders.
- TERRITORY missing values are retained rather than imputed.
- DEALSIZE is an order-line attribute; 277 of 307 orders contain multiple deal sizes.
- 2005 is partial (January–May) and is not treated as a full-year comparison.
- The public Kaggle dataset is representative of the task and is not presented as Alfido Tech internal data.

Conclusion

The analysis supports a balanced growth strategy: protect the high-contribution Classic Cars franchise and major markets, selectively scale faster-growing categories, prepare operations for the recurring Q4 peak, and reduce revenue exposure from order exceptions.